

Applied Robotics Laboratory: Conveyor Tracking

Hardware Overview and Documentation

1 Mechanical Setup

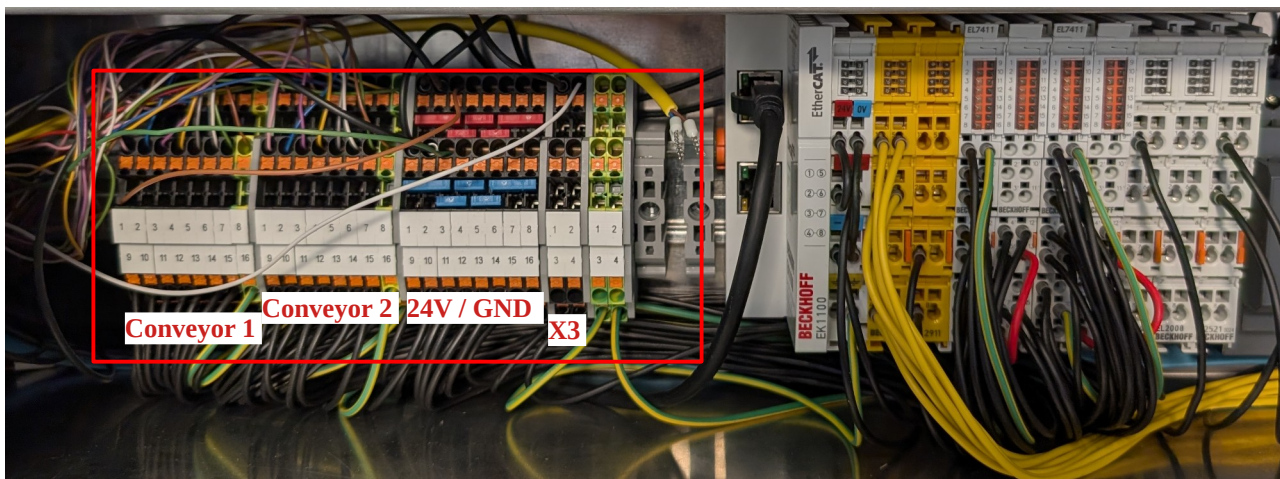
CAD is handed in separately and includes all 3D-Printed parts (STL/STEP).

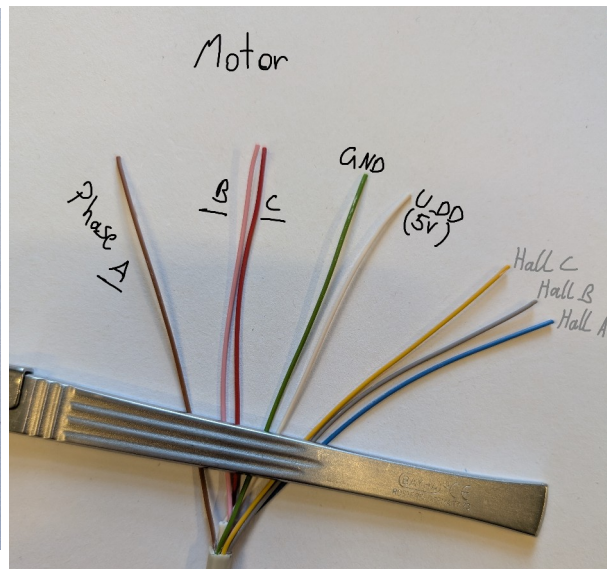
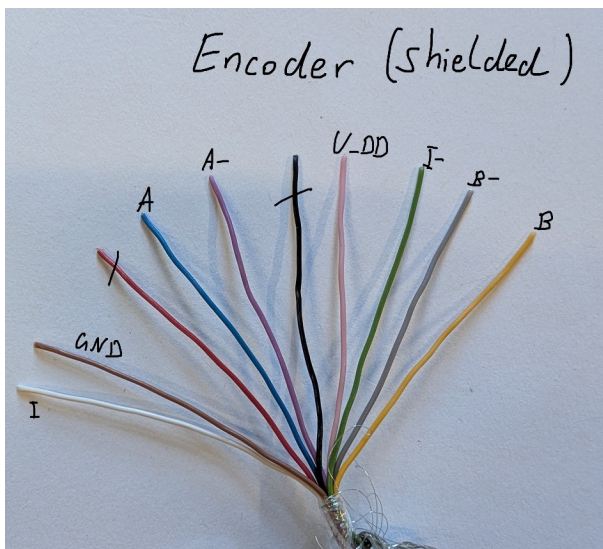
2 PLC Setup

2.1 Wiring

Component	Usage	Notes
EK1100	EtherCAT	
EL2911	Safety	
EL7411	BLDC Drive Conveyor	
EL7411	BLDC Drive Conveyor	
EL1014	Digital Input: Hall Sensor	
EL2008	Digital Outputs: R/G LED	
EL2521	Encoder Simulation Output to DSQC377	On old setup: ENC simulation via pulsing digital outputs Axis Slave of EL7411

All connections are broken out:





Encoder Channel A, B and C (or I) must all be connected to EL7411.

The Hall Sensor of the motor is not required but still has been connected.

Only minor changes have been made to the electrical system after the first EPLAN was made. It might be useful to update the EPLAN to represent the current state.

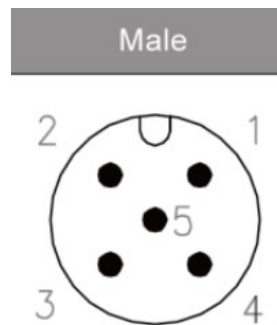
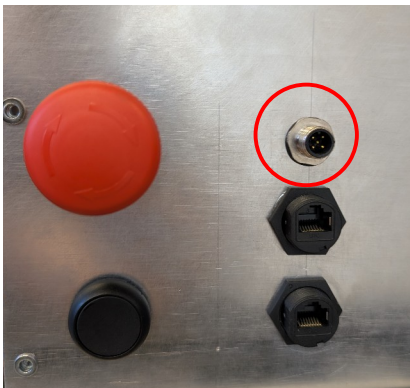
The Terminal Assignment can be seen on the next page.

			Conveyor 1			
	Motor	UDD	1			7
		Phase A	2			4°
		Phase B	3			12°
		Phase C	4			5°
		Hallsensor A	5			5
		Hallsensor B	6			6
		Hallsensor C	7			13
		GND	8			15
	Encoder	UDD	9			1
		Kanal A+	10			2
		Kanal A-	11			10
		Kanal B+	12			3
		Kanal B-	13			11
		Kanal C+	14			4
		Kanal C-	15			12
		GND	16			9
	EL7411.1	13°				8°
						EL7411.1
			Conveyor 2			
	Motor	UDD	1			7
		Phase A	2			4°
		Phase B	3			12°
		Phase C	4			5°
		Hallsensor A	5			5
		Hallsensor B	6			6
		Hallsensor C	7			13
		GND	8			15
	Encoder	UDD	9			1
		Kanal A+	10			2
		Kanal A-	11			10
		Kanal B+	12			3
		Kanal B-	13			11
		Kanal C+	14			4
		Kanal C-	15			12
		GND	16			9
	EL7411.2	13°				8°
						EL7411.2
			X10			
	24V	SPSV+	1			
		IND Sensor 24V	2			IPC 24V
			3			A4-BC1 24V
			4			EL7411.1 7°
			5			EL7411.2 7°
			6			EL2911 4°
			7			
			8			
	0V	SPSV-	9			LS GND
			10			IPC GND
			11			A4-BC1 GND
			12			EL 7411.1 15°
			13			EL 7411.2 15°
			14			EL 2911 3°
			15			
		IND Sensor GND	16			
			X20			
		IND Sensor NO	1			EL 1014 1 IND
			2			LS
			3			
			4			
			X30			
	PE		1			A4-BC1 8
			2			
			3			
			4			

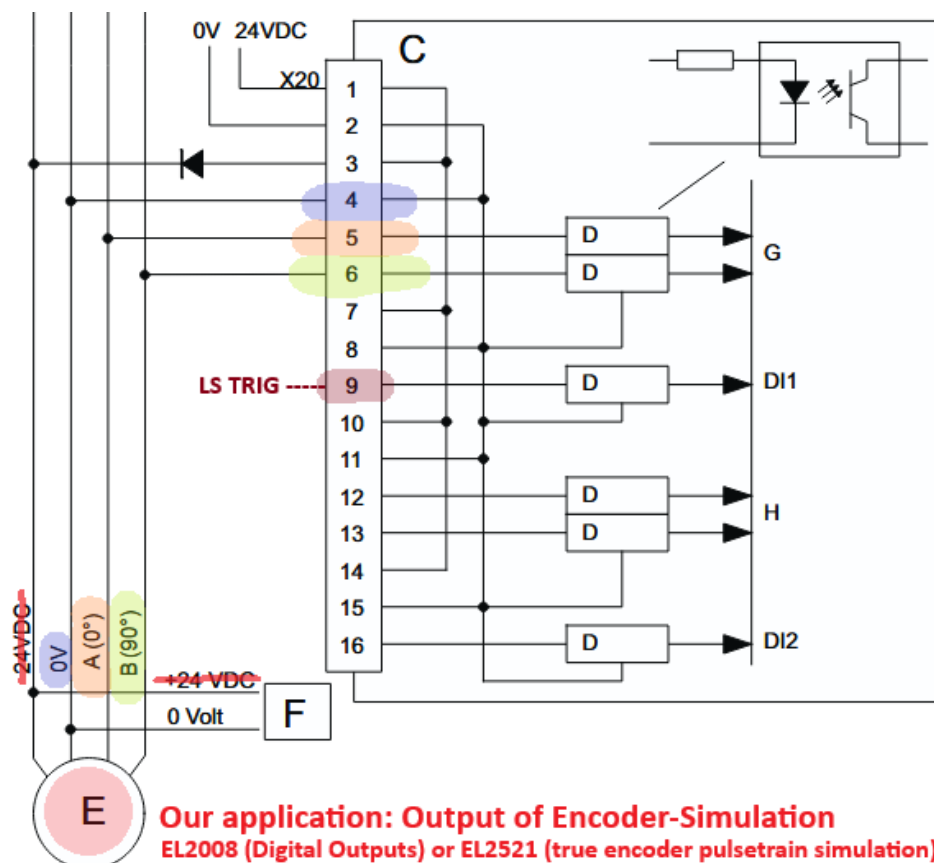
2.2 Electrical Signals PLC → Robot

M12 5-Pin Connector for ENC and TRIG signal → Conveyor Tracking

Pin	Usage
1	EncA
2	EncB
3	GND 0V
4	None
5	+LS TRIG



See ABB Document 3HAC050991-001 (Application Manual Conveyor Tracking) for detailed description on the DSQC377 card. Here is an overview of the electrical connections.



2.3 Network Connection PLC → Robot

PROFINET – use PN-Backups of Robot controllers

SPS_67 only connects to robot ...67 Wi-Fi, and SPS_68 only to ...68.

Both PLC have the static IP address 192.168.125.5 → each in the respective network. If you want to connect to the PLC via remote desktop, set your own adapter IP settings to automatic and the Wi-Fi router will give you the right address via DHCP. If you use manual addresses, use 192.168.125.XXX but not .5!

Adapter settings are automatic for PROFINET and EtherCAT – no static addresses!

2.4 Encoder Signal generation from PLC to Robot

On SPS_67, the EL2521 is used (**not yet finally working in the program**) to generate a 24V pulse (encoder) signal from the speed of the EL7411 Camera_Axis.

On SPS_68, the EL2008 Digital Output card is used to generate a 24V pulse signal with the delegated task and Main_Encoder program.

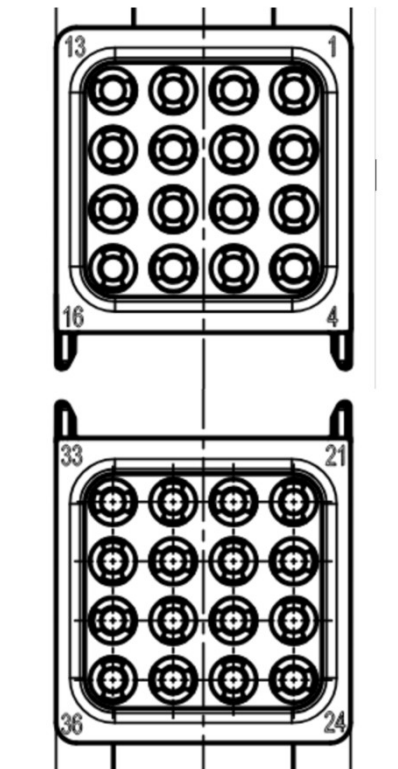
Both have the same purpose: Generate a 24V encoder signal for the DSQC377 (ABB Conveyor Tracking) to process. **The DSQC377 needs a minimum level of 18V but the encoder in the conveyors is TTL 5V.** The solution with EL2521 is the correct, recommended way to solve this issue. Using digital outputs is only a temporary workaround.

3 HARTING

On the old setup, the two modules are separate and a connection is established using the HARTING industrial connector. The “Han Gelenkrahmen Plus” frame holds 4 “Han Domino” cubes. Power transmission for the camera, power and signal for the light barrier (LS) sensor and all signals for the camera conveyor are routed via two “Han Domino HD” cubes – each with 16 pins, so a total of 32 pins is available on the HARTING connector. Conveniently, the 16-pin cubes have labels 1-16 and 21-36 for each pin, as visible on the image. The pin occupation is as follows:

HARTING Pin Occupation

Pin	Usage
1-10	Encoder (passthrough, wires see chapter 2.1)
11	+24V (for LS and Camera)
12	+Out LS (Trig)
13	GND 0V
14	-
15	-
16	-
21-28	Motor (passthrough, wires see chapter 2.1)
29-36	-

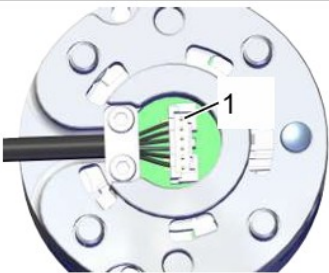


Ethernet for the COGNEX camera is passed through the HARTING connector as well.

4 SCHMALZ

Both Vacuum generators were delivered with cables attached. SCHMALZ uses JST-PA style (RS: 476-6805) connectors. The original cables were removed and a short cable for the quick-change electrical passthrough was soldered to a JST-PA connector. This was done so that the original SCHMALZ cables don't need to be cut. The wiring can be seen here:

PIN-Belegung Flanschstecker 6-polig

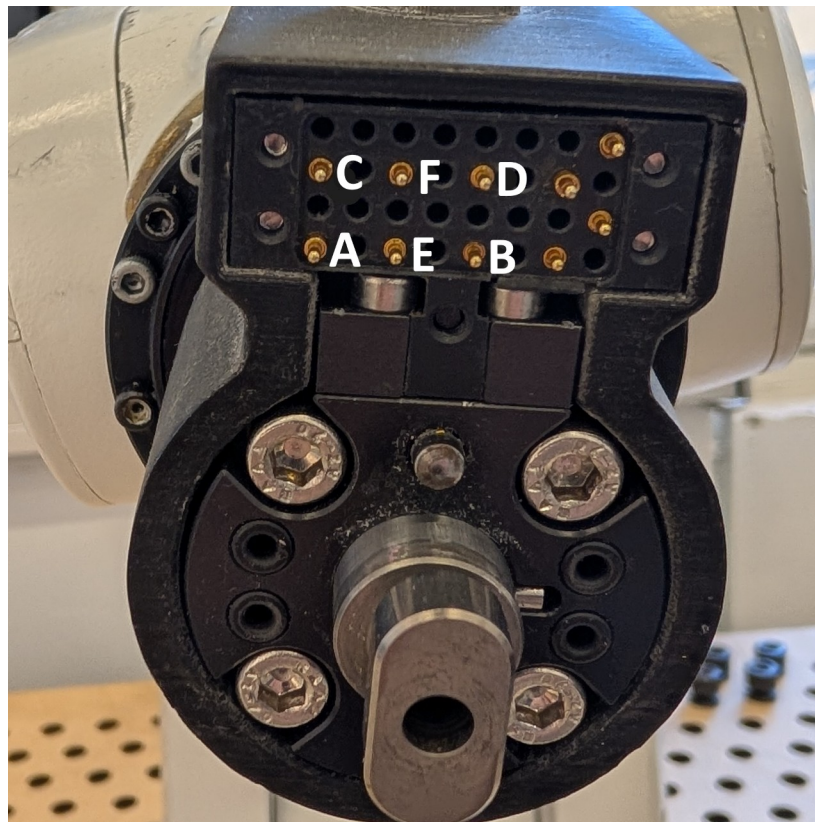
Flanschstecker	PIN	Symbol	Funktion
	1	IN2	Signaleingang „Ablegen“
	2	IN1	Signaleingang „Saugen“
	3	OUT3	Optionaler Signalausgang z.B. Freedrive aktivieren
	4	OUT2	Signalausgang "Part Present" / IO-Link
	5	GND	Masse
	6	U	Versorgungsspannung 24 V

The connections must match the pins of the quickchange system on the robot. See the yellow folder in the lab office for a detailed layout of the quickchanger.

SCHMALZ Quickchange Pins

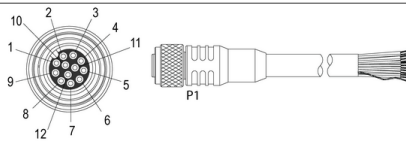
Quickchange Pin	Function
A	+24V
B	DO1 (DO07)
C	GND 0V
D	DO2 (DO08)
E	DI1 (DI07?)
F	DI2 ?

Watch out! The pins on the receiving end of the quick changer are mirrored!



5 COGNEX

The camera tower is detachable from the base. The camera is connected via a CCB-M12X12FS-05 cable to the breakout wago clips inside the camera tower.



Pin#	Signal Names	Wire Color
1	Out 2/In 2	White
2	TxD	Green
3	RxD	Pink
4	Out 3/In 3	Yellow
5	In 1	Grey
6	Common In pull to ground for PNP trigger!	Black
7	+24 VDC	Brown
8	GND	Blue
9	Common Out	Purple
10	In 0 +LS TRIG (PNP)	Red
11	Out 0	Grey/Pink
12	Out 1	Red/Blue

Important: Look closely at the label of the COGNEX cable, there are different models with different wire colors!!!

At the bottom of the camera tower, as well as at the base, an A-coded M12 8 pin connector is used to pass through the electrical connections for the camera:

COGNEX Cable → Camera Tower

Function	Cognex Cable	M12x8 Pin Cable	M12x8 Pin
+24V	Brown	Brown	2
GND 0V	Blue + Black	Blue	7
+LS TRIG	Red	Yellow	4

M12 8 pin connector matches to the camera tower base which has labeled wires and connects via Wago clips to the rest of the system from above.

Ethernet for the camera is passed through the whole system down into the Wi-Fi router of the robot! The robot uses its native network protocol for communication with the camera.

6 Bill of Materials

The following table gives an overview of important, specifically sourced components used. It does not claim to be complete and many of the components used and not listed are normally in stock at the MCI robotics lab.

Name	Usage in Component Group	Count per Setup	RS Number or Manufacturer Part Number
Festo Light Barrier	Camera Tower Base	1	SOOD-TB-R-PN
Eaton LED R/G	Part-Conveyor Cover	1	180799 M22-FLED-RG
Eaton Push Button	Main	1	M22S-D-S 216591
Eaton E-Stop Twist Release	Main	1	263467 M22-PVT
MeanWell 24V 6.25A PSU	Main	1	HDR-150-24
RS PRO RJ45-Coupler Panelmount Straight	Main	2	111-6746
RS PRO RJ45-Coupler Panelmount 90°	Camera Tower Base, Camera Tower	2	111-6747
RS PRO M12 5-pin Male Connector Panelmount	Main	1	877-1154
RS PRO M12 8-Pin Male Connector Panelmount	Camera Tower Base	1	208-0595
RS PRO M12 8-Pin Female Connector Panelmount	Camera Tower	1	208-0582
RS PRO M12 8-Pin Male Connector for Cable	Connection Cable Camera Tower	1	208-0570
RS PRO M12 8-Pin Female Connector with open Cable	Connection Cable Camera Tower	1	206-7628
Assemtech Inductive Sensor	Main	1	866-2628